

Dr Oliver Bayley

Research Engineer • ML and Embedded Systems

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PROFILE

Research engineer with a PhD in chemistry, building machine-learning coupled hardware systems. For the last three years I have designed self-driving laboratories end-to-end where Bayesian optimisation (BO) agents choose experiment conditions, custom-built hardware runs it, and automated data pipelines turn the raw instrument data into the next training point. This work has produced three platforms (see *publication list*) and industrial collaborations with *Novo Nordisk*, *Covestro* and *Symeres*.

TECHNICAL SKILLS

Machine learning: Bayesian optimisation (BoTorch, Gaussian processes; multi-objective and transfer learning), gradient boosting (XGBoost), scikit-learn (RF, SVM, KNN, PLS, MLP), PyTorch and PyTorch Lightning, hyperparameter optimisation (Optuna)

Data: pandas, NumPy, SciPy; pipelines for noisy 1D and 2D signals and time-series instrument data (chromatograms, spectra); visualisation with matplotlib, seaborn, TensorBoard

Languages: Python (primary), C++ (Arduino/embedded), Rust (bare-metal, Embassy) and Go (familiar)

Software engineering: Git, GitHub Actions (CI), pytest, Pydantic and Hydra (configuration and validation), documentation (MkDocs + github.io), GUIs (PyQt, Streamlit, tkinter)

Hardware & robotics: firmware in C++ (Arduino) and bare-metal Rust (RP2350); KiCad PCB design; Klipper/Moonraker motion control (Raspberry Pi + STM32 board); CAD (Autodesk Inventor) and FDM 3D printing

Cheminformatics & lab automation: RDKit (molecular descriptors, ECFP/FCFP fingerprints); automated data pipelines for HPLC, UV-Vis, MS and NMR

AI tools: Claude Code, Codex, Windsurf, used daily for development, refactoring and code review

EXPERIENCE

Guest Research Fellow

Jul 2026 – Present

University of Amsterdam, Noël Research Group — Amsterdam, Netherlands

- Maintaining and refactoring the RoboChem self-driving-lab codebase, preparing manuscripts, and supervising PhD students building the next generation of platforms.

Postdoctoral Research Fellow

Nov 2023 – Jun 2026

University of Amsterdam, Noël & Gargano Research Groups — Amsterdam, Netherlands

- Built and owned full stack from ML decision layer (1) to scheduling and instrument data processing (2) to the low level embedded systems and PCB circuitry (3) of three self-driving laboratories (RoboChem-Flex; RoboChem-UP; eRoboChem).
- 1. ML Layer included: BO-based optimisation for a low-data, high-noise regime where every data point is a real chemical experiment: GP surrogates, weighted multi-objective acquisition, transfer-learning hot-starts from earlier campaigns, and informed resampling of noisy results.
- 2. Scheduling layer included: Data pipelines that turn raw HPLC, UV-Vis, MS and NMR output into model-ready objectives; released ChromTroller, an open-source Python package for automated HPLC control and analysis.
- 3. Full stack beneath the scheduler included: Python orchestration and device-control software, C++ firmware, custom KiCad PCBs and 3D-printed devices (pumps, valves, electrochemical, photochemical, gas and thermal modules) at around 5% of the cost of commercial equivalents.
- Supervised PhD and MSc researchers across chemistry, hardware and software, including code review.

Postgraduate Research Assistant

Jun 2019 – Dec 2019

University of Auckland Cancer Society Research Centre (ACSRC) — Auckland, New Zealand

- Multi-gram synthesis of an anti-cancer compound for pre-clinical trials.

Earlier Research & Teaching Roles

2016 – 2019

Postgraduate Research Assistant, Victoria University of Wellington · Research Intern, Callaghan Innovation · Laboratory Demonstrator and private STEM tutor.

EDUCATION

PhD, Chemistry — University of Bristol, UK

Mar 2020 – Feb 2024

Thesis: Development of a New Class of Molecular Machine | **Passed with no corrections.**

MSc, Drug Discovery and Development — Victoria University of Wellington, NZ

Feb 2018 – May 2019

Thesis: Synthesis of Novel Pyran Fragments to Incorporate into Peloruside Analogues. | **Distinction**

BSc, Biomedical Science — Victoria University of Wellington, NZ

Feb 2015 – May 2018

Molecular Pharmacology and Medicinal Chemistry | **Dean's List.**

SELECTED PROJECTS

RoboChem platforms — Self-driving laboratories

University of Amsterdam

- **RoboChem-Flex:** low-cost modular platform; multi-objective and transfer-learning BO across thermal, photochemical and biochemical reactions. (*Nature Synthesis*, 2026).
 - The platform costs about US\$5,000, against US\$100,000 or more for comparable systems.
 - Across six case studies it matched or beat literature yield benchmarks while also providing pareto front trade-offs for other desirable objectives.
 - It cut one photoredox reaction from 5 h to 2 min at equal yield, more than doubled deuterium incorporation (71% vs ≤38%) in a photocatalytic H-to-D exchange and reached a 90% yield on a new substrate after only four iterations thanks to transfer learning strategies.
- **RoboChem-UP:** closed-loop optimisation of plastics upcycling, with automated polymer handling and polymer-distribution analysis (*ChemRxiv*, 2026).
 - The first self-driving lab built to work directly on waste polymer upcycling.
 - Achieved complete polyurethane depolymerisation with monomer recovery successful on 20-fold upscale
 - Could perform selective trifluorination of polystyrene to set functionalisation levels.
- **eRoboChem:** SDL for optimising electrolysis under drifting, noisy conditions including electrode fouling, gas evolution, flow and batch (*manuscript in preparation*).
 - Every case study benchmarked BO against literature results, human experts or DoE.
 - Raised a known literature electrosynthesis from 88% to 97% yield.
 - Achieved an 81% yield against a 68% best from an expert chemist via OFAT using the same 26-run budget while also performing a dual objective optimisation.
 - In another high-throughput batch reaction case the BO (37%) beat the DoE (27%) results

ChemRover — Molecular property prediction benchmark

github.com/ombayley/ChemRover

- Config-driven ML pipeline (Hydra, Optuna, scikit-learn, XGBoost, RDKit) predicting azobenzene absorption maxima from chemical structure and experimental conditions: 8 models (Mean, PLS, SVM, KNN, GP, RF, XGB, MLP), behind one interface, with RDKit-descriptor and ECFP/FCFP fingerprint encodings and resumable Optuna tuning.
- Best models reach $R^2 \approx 0.92$ (RMSE ≈ 17 nm) on held-out data with multi-seed learning curves showing split-to-split variance of $R^2 \approx 0.03$. Part of preliminary work for a 2026 Marsden Fund proposal.

ChemKlipper — Async control layer for a low-cost liquid-handling robot

github.com/ombayley/ChemKlipper

- Asyncio Python package (reconnecting WebSocket JSON-RPC client, pydantic configs, pytest, mypy, MkDocs on GitHub Pages) driving a Klipper / Raspberry Pi motion system. Cut command latency from ~350 ms to ~15–20 ms median through transport, firmware-buffer and CPU-isolation tuning. Built to pair with ChemRover so model predictions can be tested automatically. Part of preliminary work for a 2026 Marsden Fund proposal.

PUBLICATIONS

7. *Regnier, M.; ***Bayley, O.**; Hopsort, G.; Brunetti, A.; Savino E.; Gargano, A. and Noël, T. Autonomous Flow Electrolysis: A Self-Driving Platform for Electrochemical Reaction Optimization. *Manuscript in Preparation*
 6. *Vanzella, M.; ***Bayley, O.**; ... Noël, T. Autonomous Control of Polymer Upcycling with a Self-Driving Laboratory. *ChemRxiv*, 2026, DOI: <https://doi.org/10.26434/chemrxiv.15005284/v1>
 5. Djossou, J.; Claros, M.; **Bayley, O.** and Noël, T. Redefining synthetic efficiency: Chemical and technological shortcuts. *Chem*, 2026, DOI: <https://doi.org/10.1016/j.chempr.2026.103139>
 4. *Pilon, S.; *Savino, E.; ***Bayley, O.** M.; Vanzella, M.; ... Noël, T. A flexible and affordable self-driving laboratory for automated reaction optimization. *Nature Synthesis*, 2026, DOI: [10.1038/s44160-026-01053-0](https://doi.org/10.1038/s44160-026-01053-0)
 3. *Djossou, J.; *Pasca, F.; ... **Bayley, O.**; ... Noël, T. Radical Disconnection Logic Enables Direct Conversion of α -Amino Acids into Differentiated Vicinal Diamines. *J. Am. Chem. Soc.*, 2026, DOI: <https://doi.org/10.1021/jacs.6c16323>
 2. Kaplaneris, N.; Savino, E.; ... **Bayley, O.**; ... Noël, T. Machine Learning-Guided Discovery of Robust Conditions for Photochemical Nickel-Catalyzed Cysteine Arylation. *ChemRxiv*, 2025, DOI: [10.26434/chemrxiv-2025-zwqnt](https://doi.org/10.26434/chemrxiv-2025-zwqnt)
 1. ***Bayley, O.**; *Savino, E.; *Slattery, A. and Noël, T. Autonomous Chemistry: Navigating Self-Driving Labs in Chemical and Material Sciences. *Matter*, 2024, 7, 2382-2398, DOI: [10.1016/j.matt.2024.06.003](https://doi.org/10.1016/j.matt.2024.06.003)
- * Equal contribution.

BOOK CHAPTERS

1. T. Noel, E. Savino, **O. Bayley**, M. Vanzella, S. Pilon, in *Automation in Drug Discovery*, M. A. Pericas, J. Alcazar, S. Canellas, J. E. Gomez Pulido, Eds., *Wiley* **2026**.

SELECTED ACHIEVEMENTS

Five best talk / best poster awards, 2023–2026 (incl. RSC 27th International Symposium, Oxford; ECR MASC, Francis Crick Institute; Flow Chemistry Europe, Málaga) · Invited Talks 2024-2026 (incl. Big Chem NL, Groningen; UvA- Max Planck Symposium) · Merck Compound Challenge, top 15 of 243 teams (2021) · Te Herenga Waka Master's by Thesis Scholarship (2018) · Ngāti Kahungunu Iwi Scholarship (2016)

REFERENCES

Available on Request